



IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

Knowledge Based Systems

Semester I TA 2026/2027





Outline

- Review:
 - What is AI
 - Intelligent Agent: PEAS, Property of Task Environment
 - Local Search
 - Adversarial Search
 - Constraints Satisfaction Problem (CSP)
- Introduction to Knowledge Based System (KBS)
 - What and Why
 - Knowledge Based Agent
 - Architecture of KBS
 - Example of KBS
 - Knowledge Representations
 - Knowledge Acquisition



Knowledge-based System (KBS): What

Apply knowledge
in solving problem

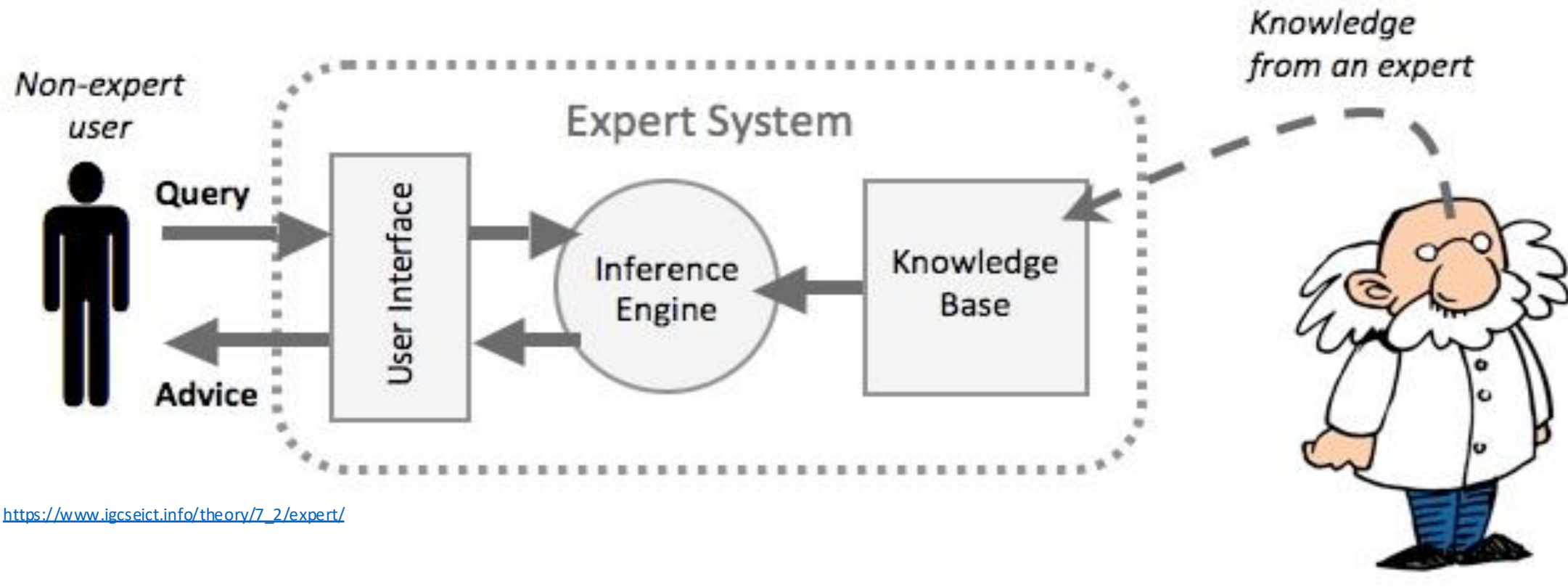
Reconstruct expertise and
reasoning capabilities of
qualified specialists within
limited domains

Logical reasoning





Knowledge-based System $\neq$ Expert System



https://www.igcseict.info/theory/7_2/expert/





Knowledge-based System (KBS): Why

Approach in
developing AI agent

Logical reasoning:
thinking rationally

Template-based
pattern recognition

Statistical-based
pattern recognition

Structural/syntactic
pattern recognition

Deep learning-based
pattern recognition



KBS

Conventional
Program

Ill-structured problem

Well-structured problem

Expert determine actions,
but execution order by
interpreter

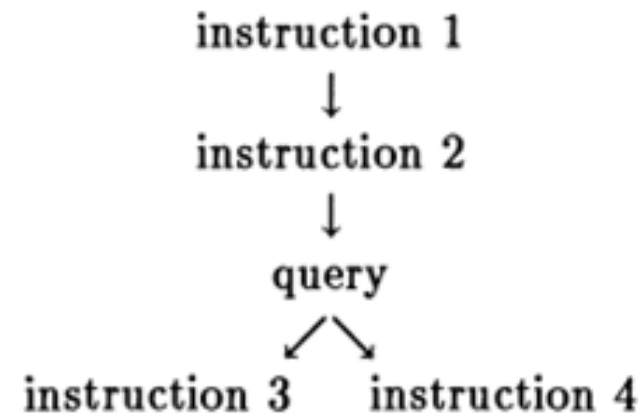
Programmer determines
actions and execution
order

Problem solving method +
domain knowledge + data

Algorithm + data



1. Instruction-based programming style:
program = sequence of instructions and queries



The programmer determines what is done and in what order it is done.

2. Rule-based programming style:
program = set of rules and rule interpreters

Rule 1: If situation X1, then action Y1.

Rule 2: If situation X2, then action Y2.

Rule 3: If situation X3, then action Y3.

The expert determines what is done, and the rule interpreter determines the order.

Puppe, F. (1993). *Systematic Introduction to Expert Systems*. Springer, Berlin, Heidelberg.



Problem Characteristics

Well-formed problem

Exact / certain solution

Explicit goal

Explicit operator

Ill-structured problem

Uncertain solution

Undefined goal

Unknown operator



Summary

What is KBS

KBS $\neq$ ES

Why KBS

KBS vs
conventional

Reasoning in Knowledge-based
Agent





Knowledge-based Agent

Fundamental properties of logical reasoning

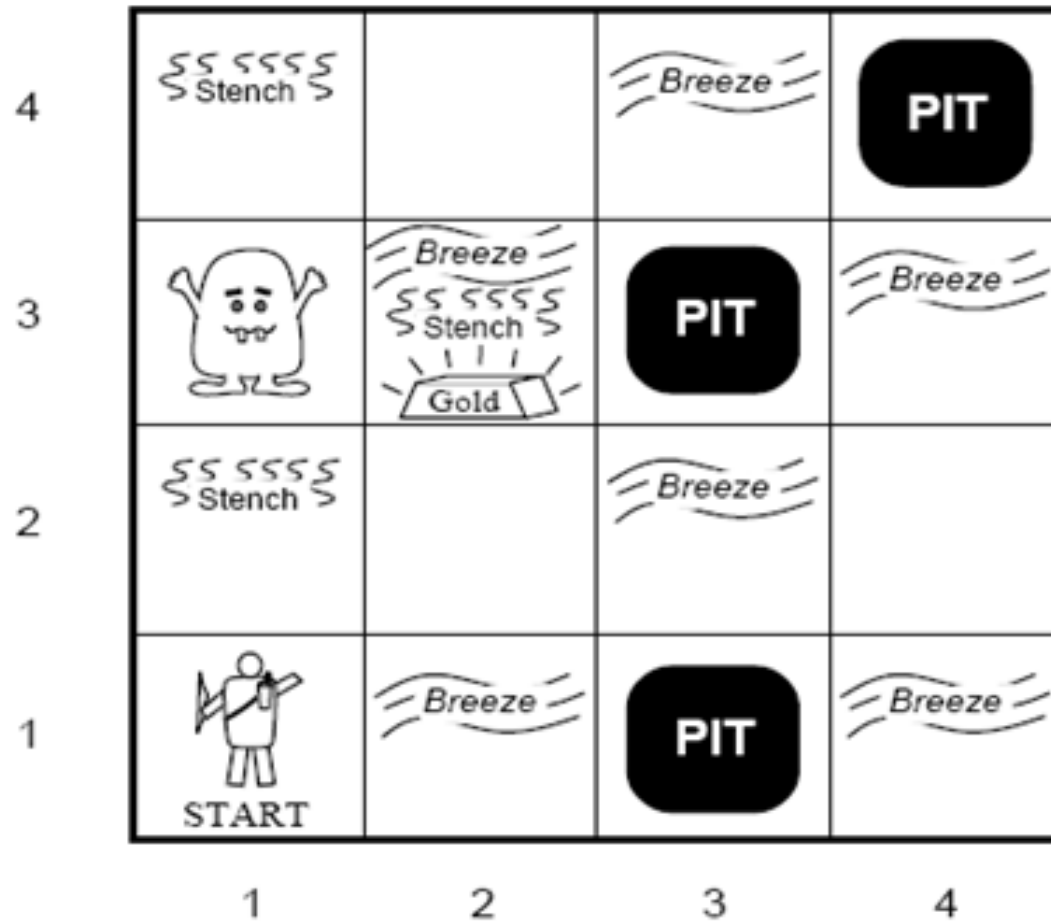
In each step, the agent draws a conclusion from available information

Conclusion is **guaranteed** to be correct if the available information is correct





Wumpus World



Performance Measure: gold +1000, death -1000, -1 per step, -10 for using the arrow

Environment: cave, rooms, Wumpus, gold

Actuators: motor to move Left, Right, Forward, hands to Grab, Release, and Shoot arrow

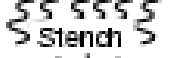
Sensors: sensor to capture [Stench, Breeze, Glitter, Bump, Scream]

11





Exploring a wumpus world

4	 Stench		 Breeze	
3	 Wumpus	 Breeze  Stench  Gold		 Breeze
2	 Stench		 Breeze	
1	 START	 Breeze		 Breeze
	1	2	3	4

Percept [1,1] : [None, None, None, None, None]
No stench in [1,1] : No wumpus in [1,2] and [2,1]
No breeze in [1,1]: No pit in [1,2] and [2,1]
Action: forward to [2,1]

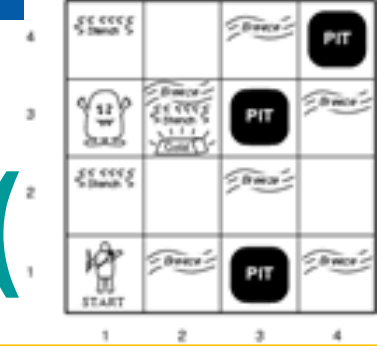
OK			
OK 	OK		

[1,1] : OK (safe)





Exploring a wumpus world (



1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

(a)

A
B
G
OK
P
S
V
W

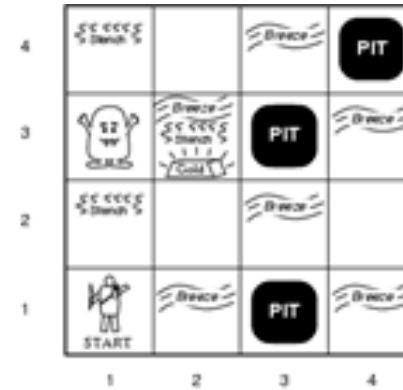
Percept [2, 1] : [None, Breeze, None, None, None]
No stench in [2,1] : No wumpus in [3,1] and [2,2]
Breeze in [2,1]: there must be a pit in [3,1] or [2,2]
Set action: go back to [1,1] and forward to [1,2]

1,2	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

(b)



Exploring a wumpus world (3)



1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

Percept [1,2] : [Stench, None, None, None, None]
Stench in [1,2] : a wumpus in [1,3] or [2,2] or [1,1]
No wumpus in [1,1] and No stench in [2,1]

❓ **wumpus in [1,3]**

No breeze in [1,2]: No pit in [1,3] and [2,2]

❓ **pit in [3,1] and [2,2] OK**

Set action: go to [2,2]



Exploring a wumpus world (



1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

Percept [2,2] : [None, None, None, None, None]
No stench in [2,2] : No wumpus in [2,3] and [3,2]
No breeze in [2,2]: No pit in [2,3] and [3,2]
Set action: go to [2,3]

Percept [2,3]: [Stench, Breeze, Glitter, None, None]
Action: Grab





Generic Knowledge-based Agent

function KB-AGENT(*percept*) **returns** an *action*
persistent: *KB*, a knowledge base
t, a counter, initially 0, indicating time

TELL(*KB*, MAKE-PERCEPT-SENTENCE(*percept*, *t*))

{assert percept}

action $\leftarrow$ ASK(*KB*, MAKE-ACTION-QUERY(*t*))

{reasoning}

TELL(*KB*, MAKE-ACTION-SENTENCE(*action*, *t*))

{assert action}

t $\leftarrow$ *t* + 1

return *action*





Knowledge-based Agent Development

Starting with an empty knowledge-base

The designers have no idea about the solution

Agent designer can TELL sentences one by one

The designers cannot anticipate all possible situations

agent knows how to operate in its environment

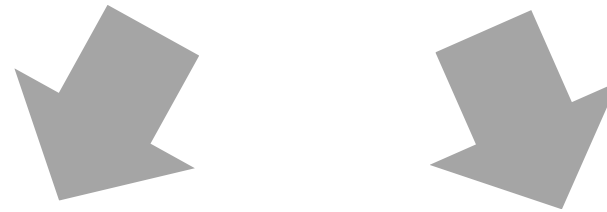
The designers cannot anticipate all changes over time





Knowledge Representation

A language (to represent knowledge/ information)
a set of syntactic and semantic conventions that makes it possible to describe things, and a way of manipulating expression in language



Syntax: a description of what you're allowed to write down, what the expressions are, that are legal in a language.



Semantic: which is some story about what those expressions mean.

18





Requirements of Knowledge Representation

No contradiction

Each symbol must be unique

Explain certain objects, relations and attributes

Efficient manipulation in computer system

Production rules

Semantic networks dan frames





Selecting Knowledge Representation

Suitable for
problem domain

- Decision tree for classification
- Skeletal construction for construction
- Rule for all problem domain

Suitable for the
tasks (inference)

- Decision tree including interview process
- Probability model for decision with uncertainty

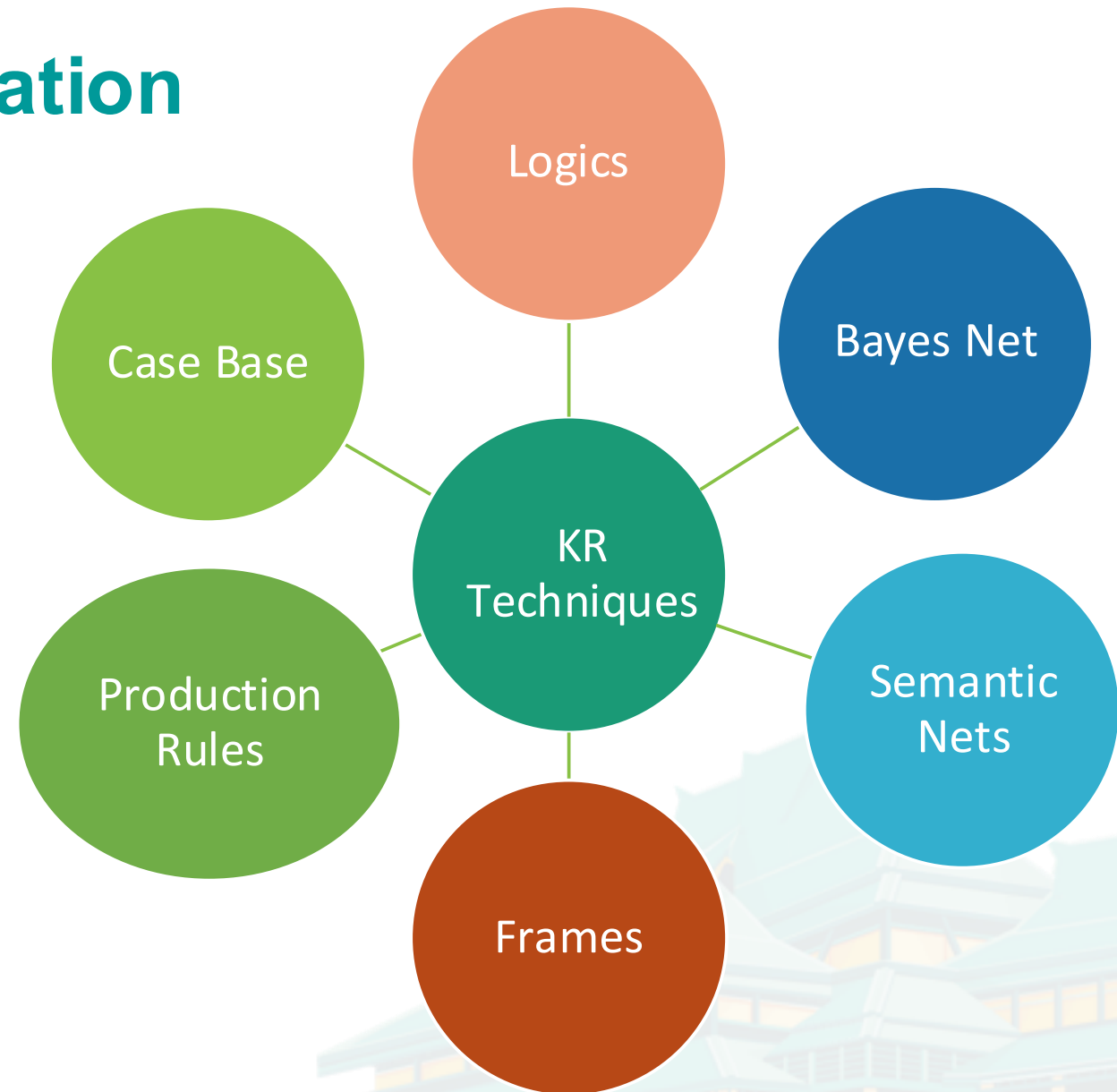
Suitable for users
(man or machine)

- Semantic network for user
- rule for machine





Variation of Knowledge Representation



Propositional and Predicate Logic

- Sistem berbasis pengetahuan yang menggunakan **logika proposisional atau predikat** untuk merepresentasikan pengetahuan umumnya melibatkan aturan IF-THEN, hubungan antar objek dan kuantifikasi objek.
- Contoh:
 - a. Smart Building Control System: menggunakan sensor untuk mendeteksi kondisi lingkungan dan membuat keputusan otomatis berdasarkan aturan logika proposisional. Jika sensor mendeteksi bahwa tidak ada orang di ruangan, lampu akan dimatikan secara otomatis.
 - b. Automated Theorem Proving: Menggunakan logika predikat untuk membuktikan validitas teorema matematika dengan cara otomatis, seperti dalam program **Prolog** yang dapat menjawab pertanyaan-pertanyaan logis berbasis fakta dan aturan yang sudah ada.





Semantic Networks (Ontology)



- **Semantic Networks** merepresentasikan pengetahuan dalam bentuk graf, di mana konsep atau objek diwakili sebagai node, dan hubungan di antara konsep tersebut diwakili sebagai edge.
- Contoh:
 - a. Google Knowledge Graph: untuk pencarian berbasis semantik,
 - b. PayPal Fraud Detection System: memanfaatkan ontologi untuk memahami pola transaksi yang berpotensi fraud. Sistem ini menganalisis relasi antara akun, lokasi, jenis transaksi, dan sejarah pembayaran untuk mendeteksi aktivitas yang mencurigakan.
 - c. Penelitian medis: menemukan hubungan baru antara penyakit, gen, dan obat-obatan, yang dapat mendukung penelitian pengobatan baru.



Frames

hotel room
specialization of: room
location: hotel
contains: (hotel chair hotel phone hotel bed)

- **Frames** adalah struktur data yang digunakan untuk merepresentasikan pengetahuan dalam bentuk slot (tempat penyimpanan informasi) dan filler (nilai atau informasi yang mengisi slot).

- Contoh:



- a. Chatbot dan Asisten Virtual: menggunakan frame untuk memahami konteks pertanyaan pengguna dan menjawab sesuai konteks.
- b. Robotic Navigation Systems: frames digunakan untuk merepresentasikan pengetahuan tentang lingkungan robot dan tindakan yang bisa diambil oleh robot, misalnya robot pembersih, robot industri manufaktur.



Production Rule



- **Production rule** adalah representasi pengetahuan dalam bentuk aturan berbasis kondisi dan aksi, atau sering disebut dengan **if-then rules**.
- Contoh:
 - MYCIN:** Sistem pakar medis MYCIN yang dikembangkan pada tahun 1970-an untuk mendiagnosis infeksi bakteri menggunakan production rules. MYCIN bekerja dengan aturan-aturan seperti "If pasien memiliki gejala X dan hasil tes Y, then infeksi bakteri Z terindikasi."
 - Snort:** Snort adalah salah satu sistem deteksi intrusi open-source yang populer dan menggunakan production rules untuk mendeteksi aktivitas mencurigakan atau serangan dalam jaringan komputer. Aturan-aturan ini dapat disesuaikan untuk mendeteksi berbagai jenis ancaman seperti serangan DDoS, brute force, atau malware.



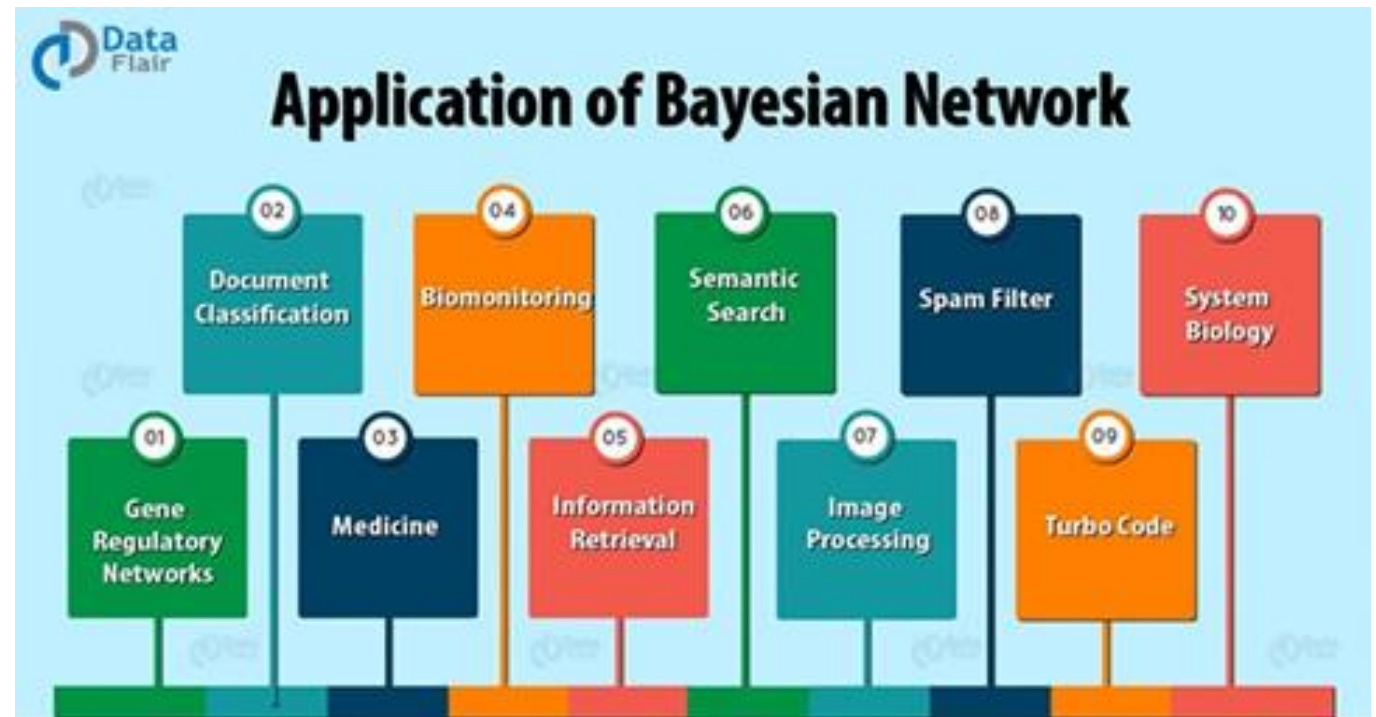
Case-Based Reasoning



- **Case-Based Reasoning (CBR)** adalah metode representasi pengetahuan di mana solusi dari masalah-masalah sebelumnya (kasus) digunakan kembali untuk memecahkan masalah baru yang serupa. Penalaran ini bergantung pada basis data kasus yang mencakup solusi atau keputusan yang diambil dalam skenario sebelumnya.
- Contoh:
 - a. **Case-Based Radiology System:** Beberapa sistem radiologi berbasis CBR membantu ahli radiologi untuk mendiagnosis gambar medis berdasarkan kasus-kasus gambar yang telah didiagnosis sebelumnya, seperti mendeteksi kanker atau kerusakan organ.
 - b. **HelpDesk Systems:** Banyak perusahaan teknologi menggunakan sistem CBR untuk memberikan dukungan teknis kepada pelanggan. Saat pelanggan melaporkan masalah teknis, sistem CBR mengacu pada kasus-kasus teknis yang telah dipecahkan sebelumnya dan menyarankan solusi.



Bayes Network



Source: <https://data-flair.training/blogs/bayesian-network-applications/>

- Reasoning under uncertainty
- Represent uncertainty not only by disjunction (Logic), but also likelihood (probability)



Summary

Logical
reasoning

Reasoning in
Wumpus
world

Generic
knowledge-
based agent

KB agent
development

Knowledge
representation

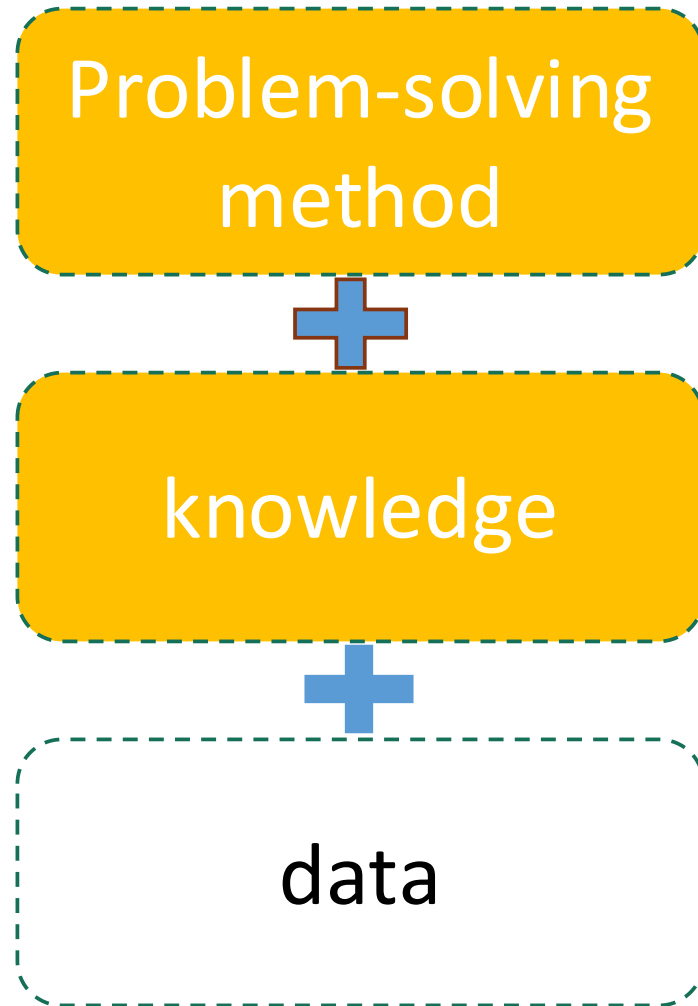
KR:
requirement &
selection

KBS Architecture





Knowledge-based System



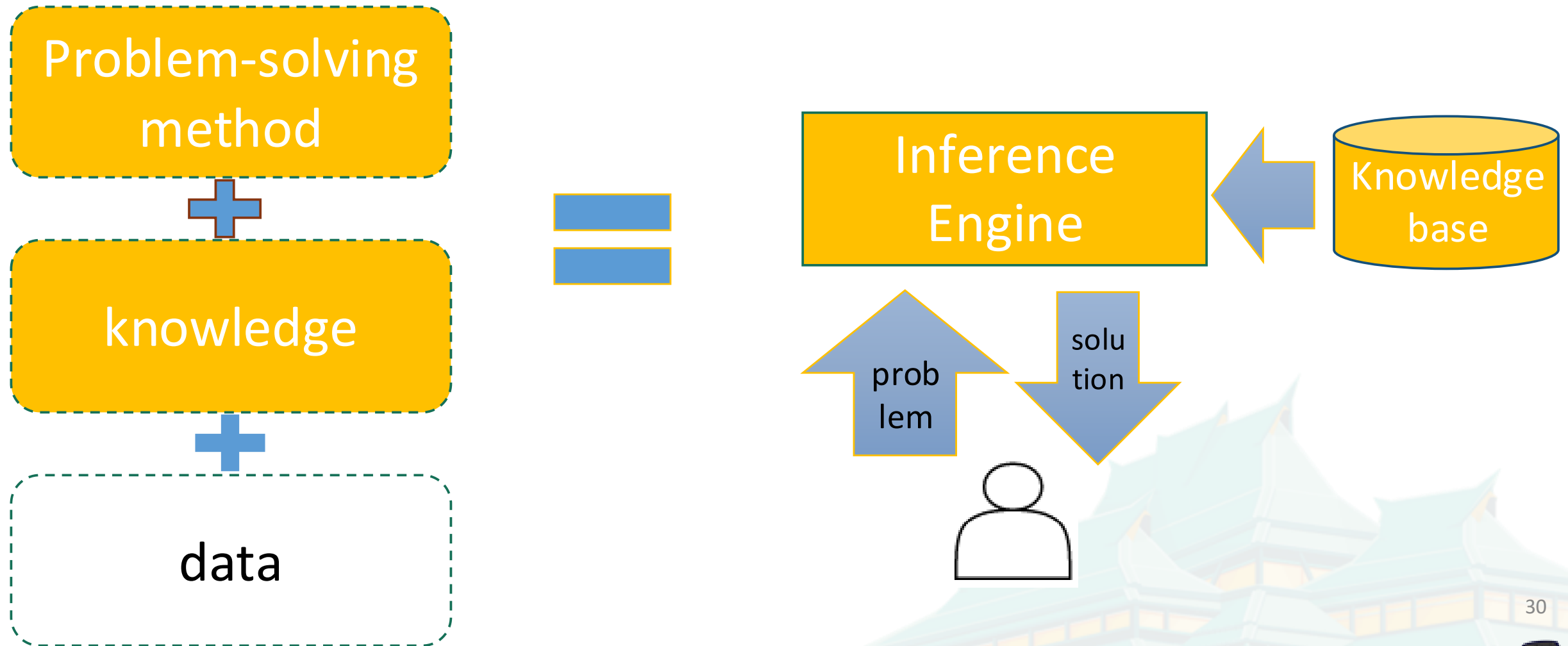
Problem-solving method is an algorithm which determines how domain-specific knowledge is used for solving problems (Puppe, 1993)

Example: knowledge reasoning methods (e.g. forward chaining for rule), general procedures (e.g. partially order plan)



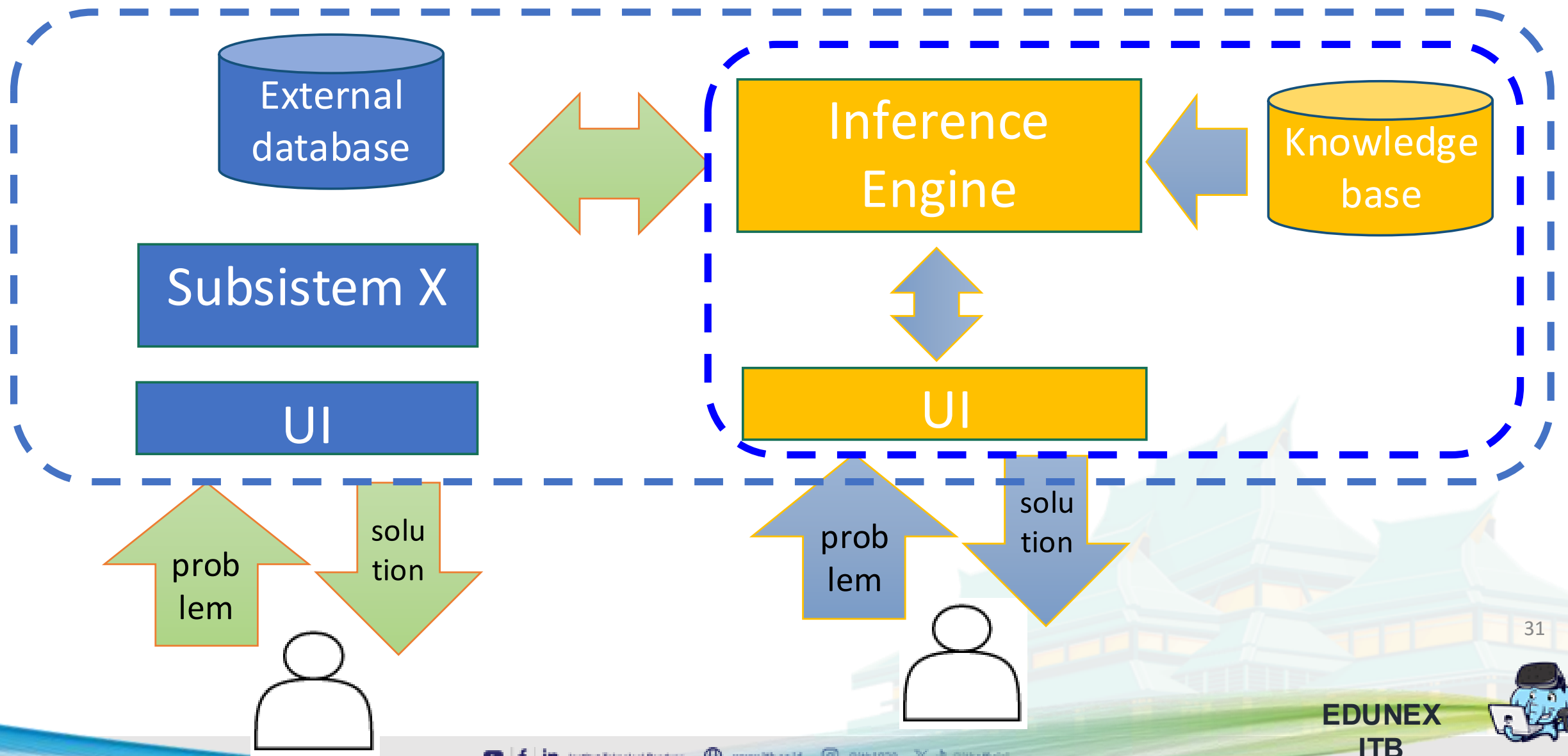


Knowledge-based System: Terminology





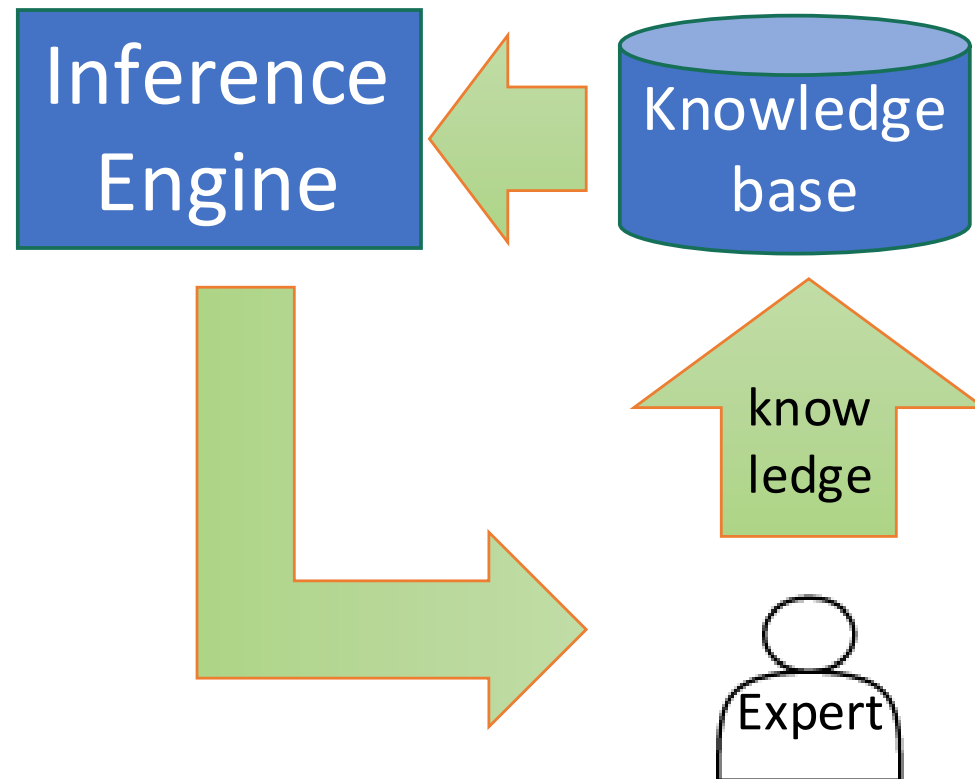
Interactive vs Embedded KBS



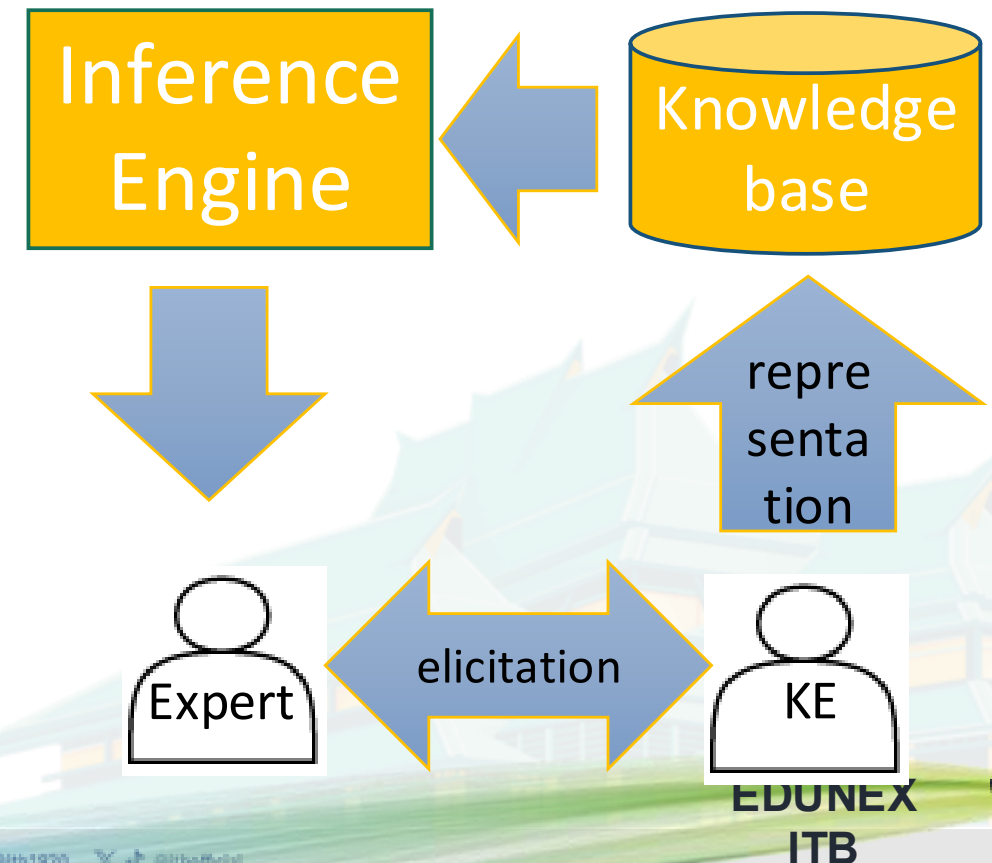


Knowledge Acquisition

Direct
knowledge acquisition

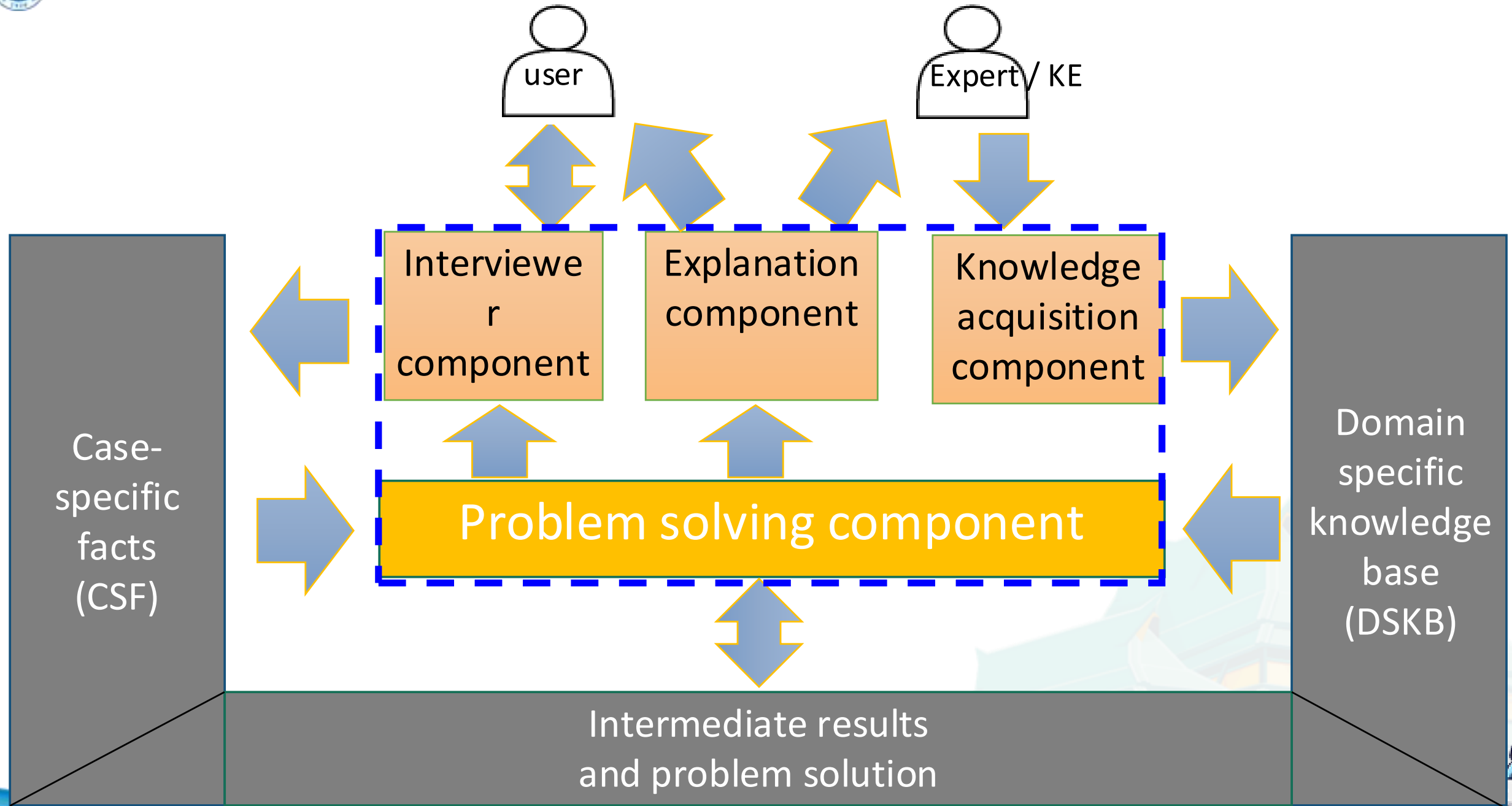


Indirect
knowledge acquisition



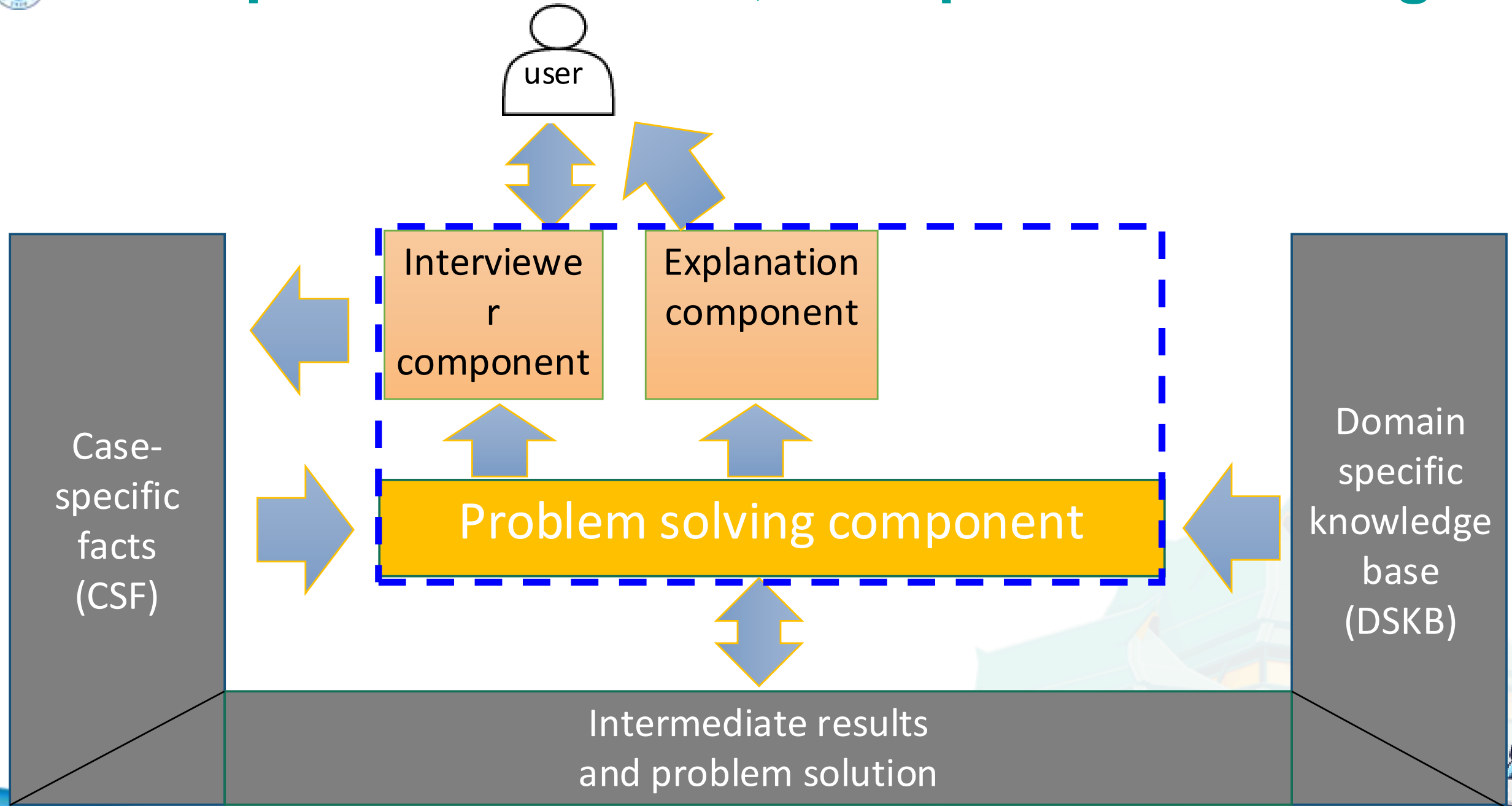


General Architecture of KBS



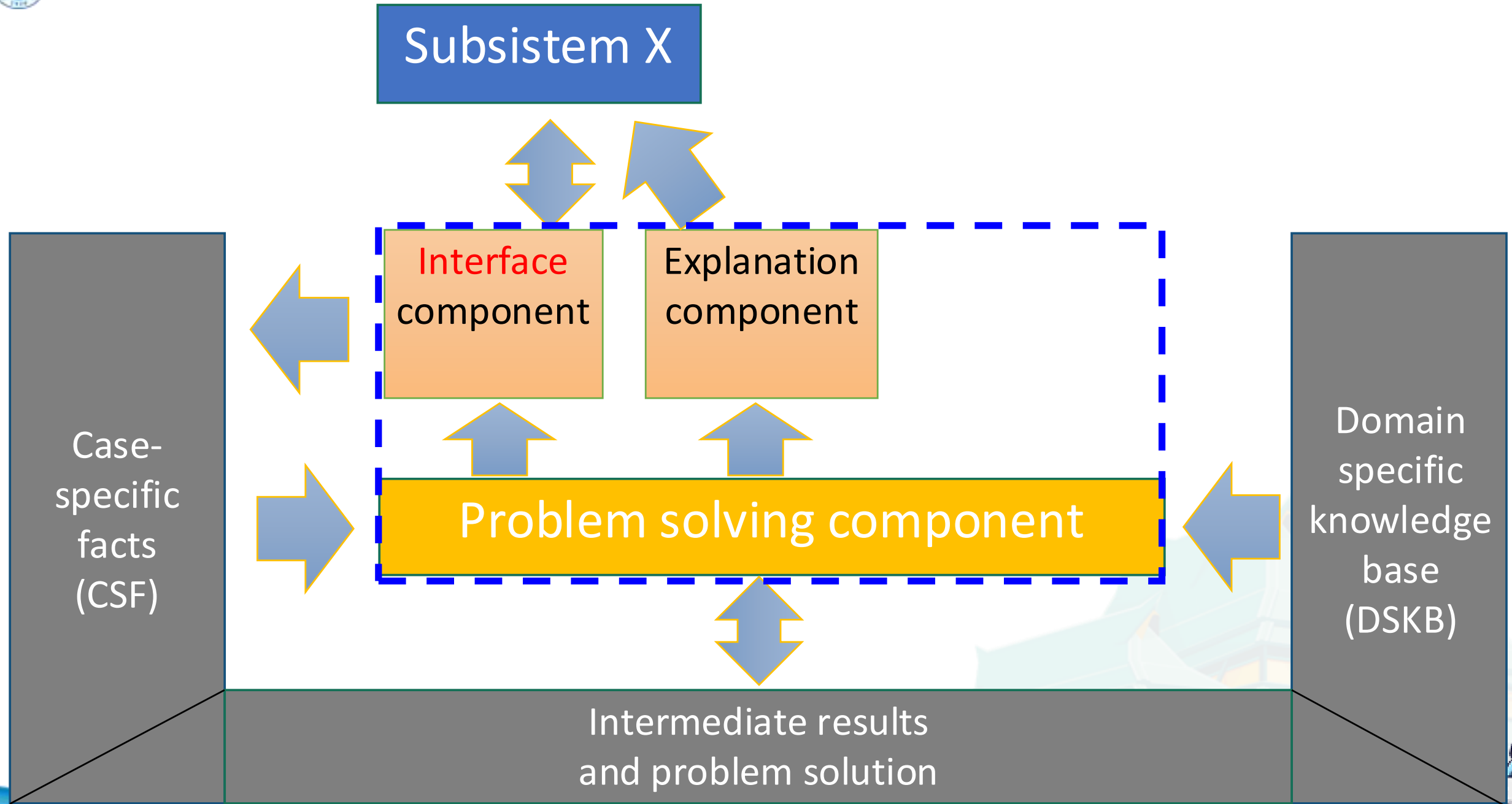


Example 1: Interactive, No Update Knowledge



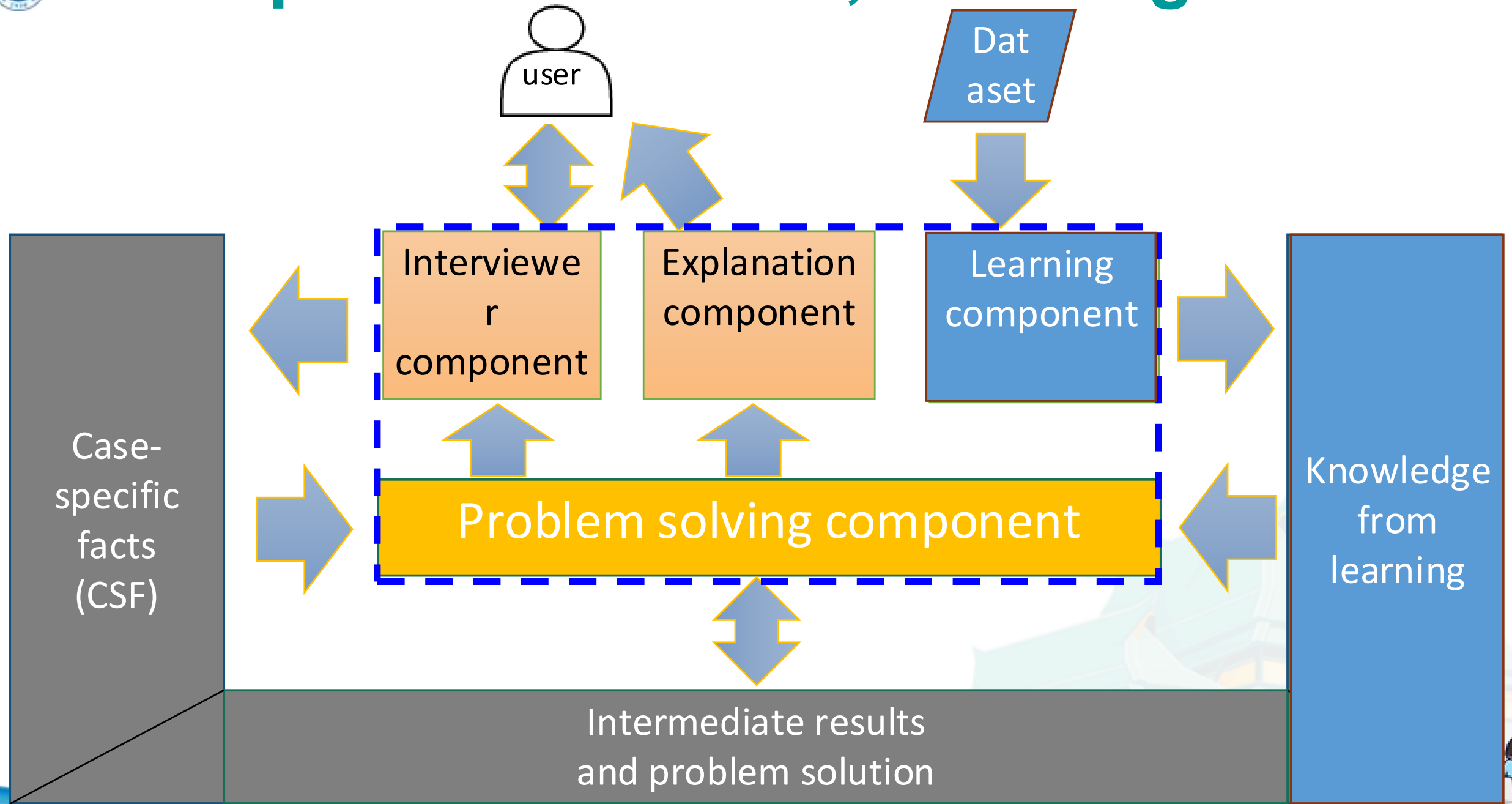


Example 2: Embedded, No Update Knowledge





Example 3: Interactive, Learning





Summary

KBS=PSM+
knowledge+
data

Inference
Engine,
Knowledge base

Knowledge
Acquisition

General
Architecture of
KBS

Examples of
Architecture



KBS Applications

- **Kesehatan:** BAL2000, LISA, ISABEL, CTSHIV, DxPlain, MedWeaver, The Analyst, FuzzyFluid, Casnet, PUFF, Centaur, EasyDiagnosis, CLEM, VIE-PNN
- **Lingkungan:** ESS-WWTP, CREWS, CORMIX, HITERM, GCES, Oncologic
- **Jaringan:** NIDES, AudES, eXpert-BSM, Expert Advisor, Online ES (listrik)
- **ITS:** ActiveMath, TEST, ELM-ART, SID2002 Math ES, Chest
- **Komputer/HW:** DART, PEARL, PDAmum
- **Manajemen:** DXMAS, CESA, FINEVA
- **Permainan:** FRES, Rogomatic
- **Geologi:** PROSPECTOR II, DAS
- **Pertanian:** EXSEL, HABES, DSS4Ag
- **Biologi:** RIH, PSORTb
- **NASA:** Weather ES, SHINE
- **Lainnya:** TTA (teroris), ACAS-PRO (kartu kredit), USLIMITS 2, CATD-RT, HWYCON, SHYSTER (hukum)





EasyDiagnosis Medical Expert System

Ads by Google Data Data Privacy Policy Data Base Modeling Visual Data Analysis USB Data Protection

EasyDiagnosis
MatheMEDics®

Headache Questions

Required: Age Sex

1. Which of the following best describes your headache?

- ☐ A. I've had them for years
- ☐ B. They started in the last few weeks or months
- ☐ C. They began recently, within a day or days
- ☐ D. Unknown/not applicable

2. Which of the following best describes location of your headaches?

- ☐ A. Occurs mainly in the back of the head or neck, and/or temples
- ☐ B. Starts on one side of the head and becomes throbbing
- ☐ C. Occurs in the frontal region
- ☐ D. Is located mainly in the eye or one side of the face
- ☐ E. More than one of above
- ☐ F. None of above
- ☐ G. Unknown/not applicable

Headache Results

Click on any disease for a description.

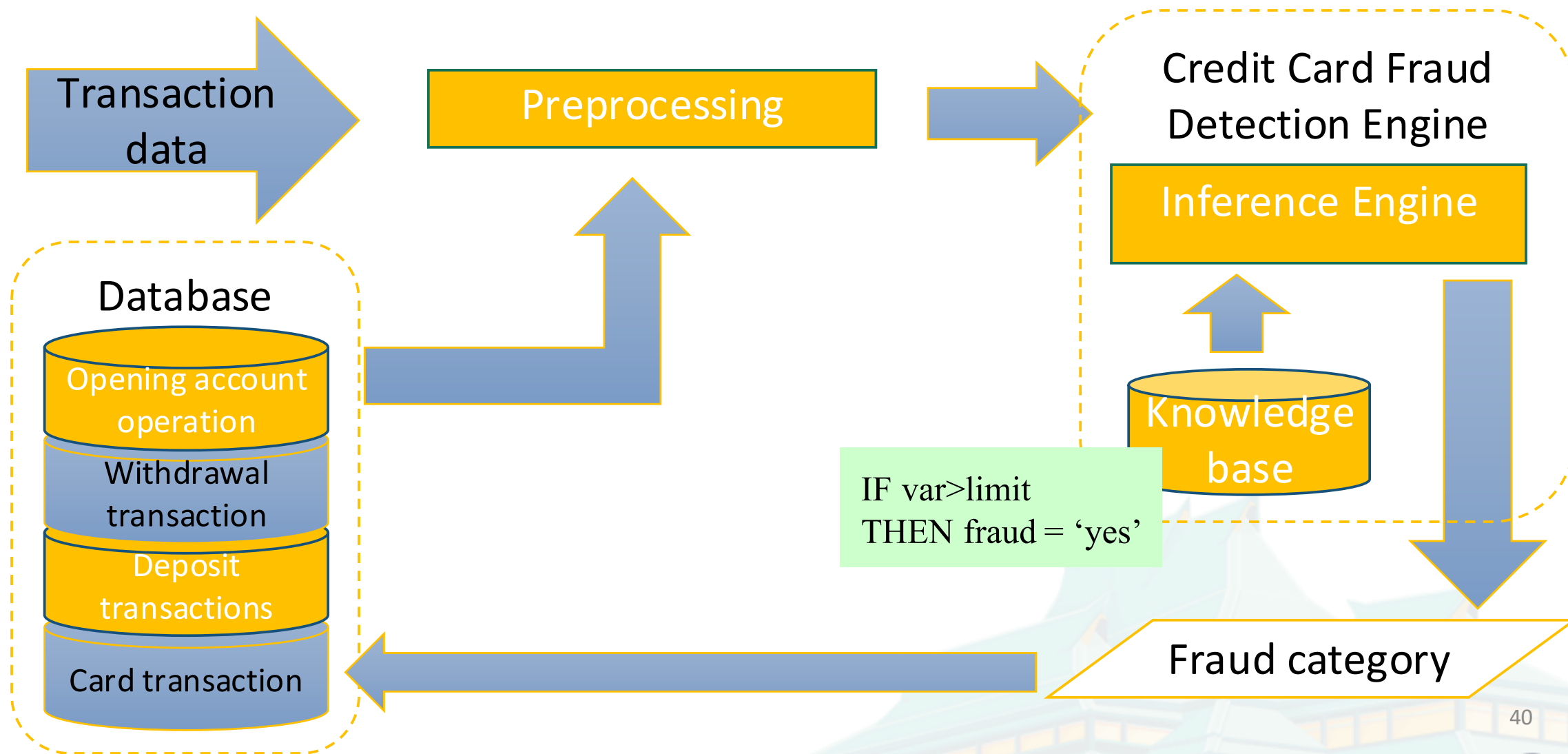
Condition/Disease	Estimated probability
Migraine	46%
Cluster Headache	30%
Temporal Arteritis	23%
Miscellaneous or Benign Headache	0.7%
Brain Tumor and Increased Intracranial Pressure	0.3%
Stroke	0.1%
Tension Headache	0.1%
Frontal Sinus Headache	0.1%
Post-traumatic Headache	0.1%
Headache-High Blood Pressure	0.1%
Headache-Meningeal Infection	0.1%
Rebound Headache	0.1%

What do these probabilities mean?





Credit Card Fraud Detection



Ogwueleka, F. N. (2011). Data mining application in credit card fraud detection system. *Journal of Engineering Science and Technology*, 6(3), 303-310.



Green Chemistry Expert System (GCES)

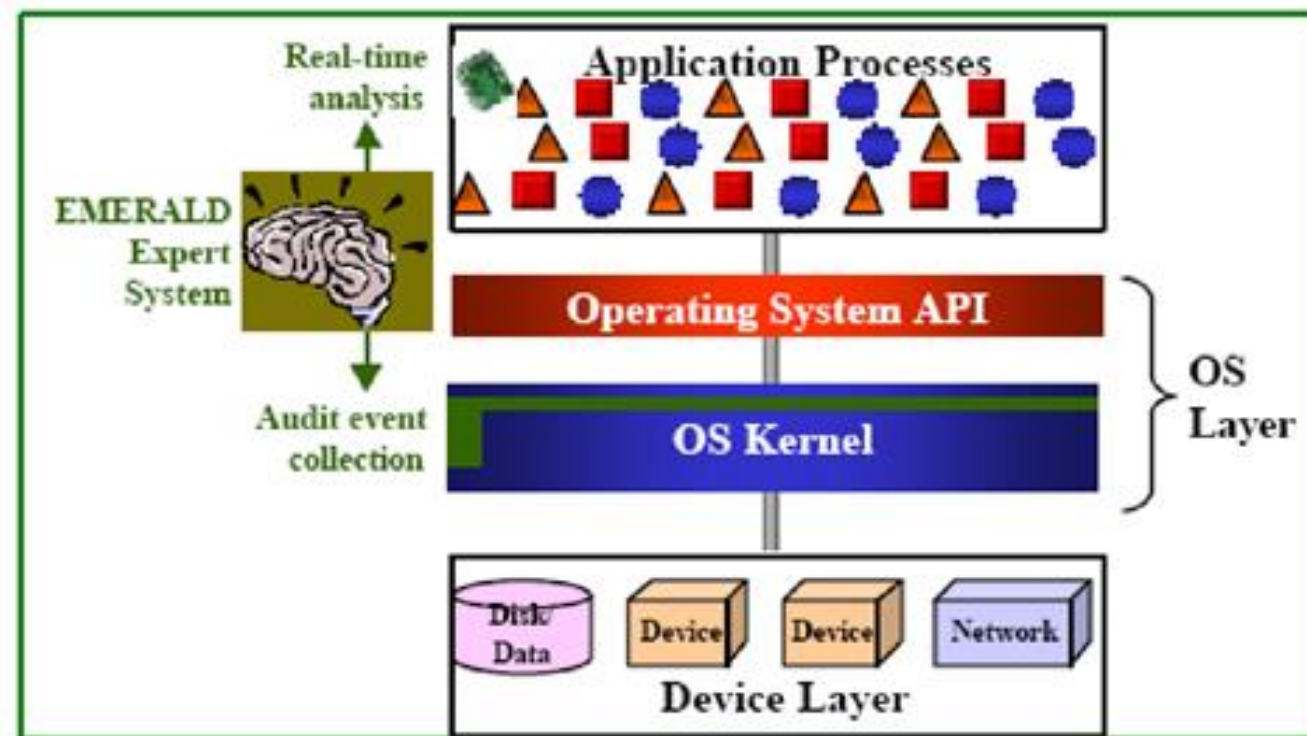
- Developer: EPA (*Environmental Protection Agency*) Amerika Serikat
 - MS Access, DBMS
- untuk menilai substansi yang berbahaya dalam reaksi kimia sehingga polusi dapat dicegah
- <http://www.epa.gov/greenchemistry/pubs/gces.html>





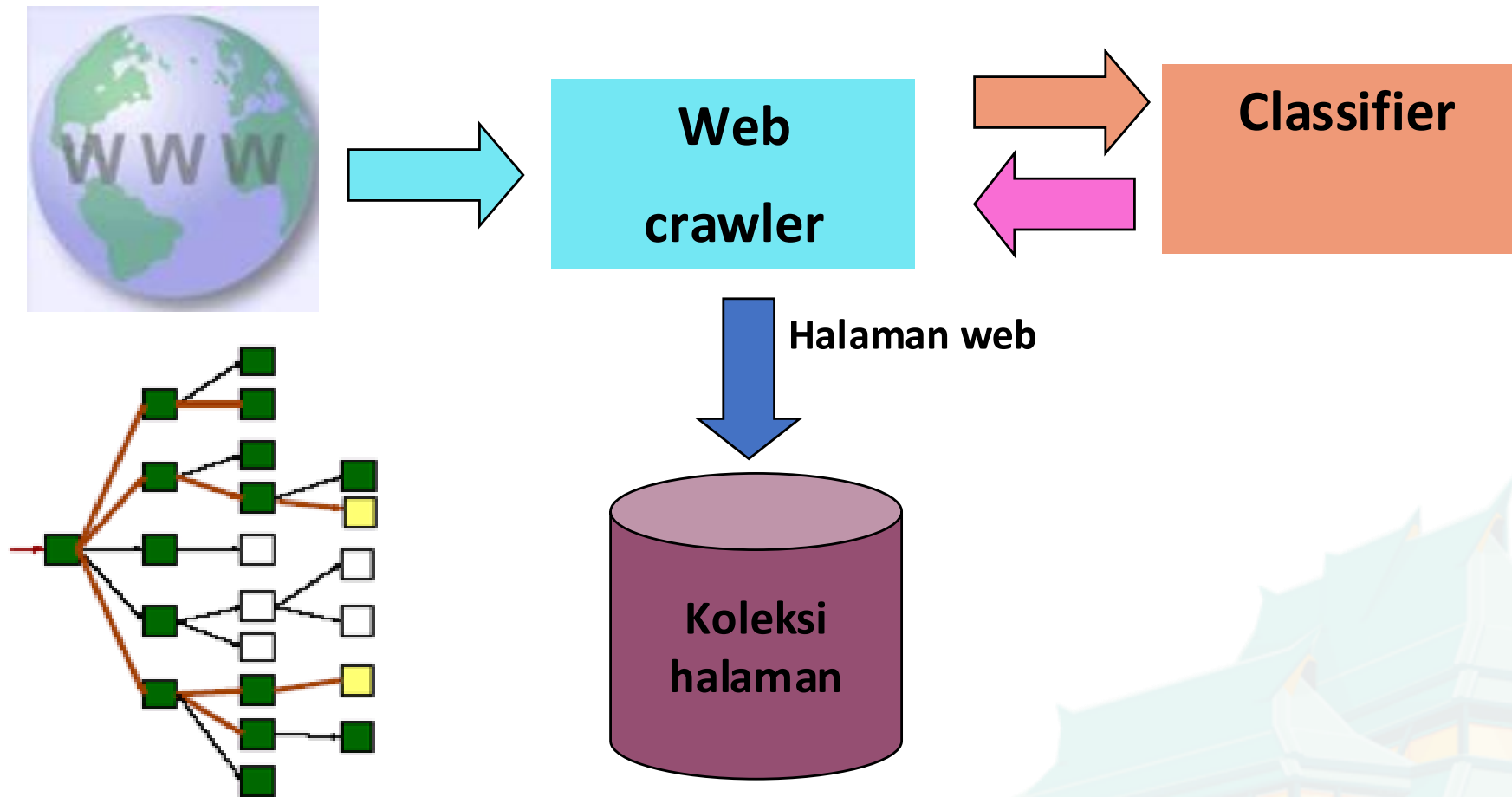
eXpert-BSM

- Intrusion Detection Solution for Sun Solaris
- Output: hasil analisis dan alert adanya intrusi pada audit trail dari Sun Solaris
- Sub sistem Emerald ES





Focused Crawler Domain X





Vertical Search Engine

Nile Guide
Plan Faster, Travel Smarter

Welcome, retdkrald | Sign Out
Trip Planner | My Trips
My Account | My Itineraries
Control Panel

San Francisco Bay Area
Change Destination

Overview | Trips | See & Do | Food | **Lodging** | Nightlife

Results: 1-15 from 690

Preferences: ☐ Business ☐ Boutique ☐ Resort ☐ Family ☐ Brand Name ☐ Hip

Neighborhoods: [All]

Types: [All]

Cost: [Slider from 5 to 55000]

Start: [Slider from 1 to 5]

Keywords search: [Field] [Find]

Nile Recommendations for San Francisco Bay Area Lodging

Results Sorted by: Top Rated Distance From: Union Square

Added to Trip	Image	Name	Address	Type	User Rating	Distance	Check Rates
☐		1801 Inn	Queen Anne Victorian	Napa	User Rating: 5.0	35.7 mi	
☐		Hotel Rex	Novel worthy	Union Square	User Rating: 4.7	0.1 mi	Check Rates
☐		Omni San Francisco Hotel	Renaissance-style hotel	Nob Hill	User Rating: 4.7	0.4 mi	Check Rates
☐		Hotel Carlton	Classic simplicity	Downtown/Financial District	User Rating: 4.6	0.5 mi	Check Rates
☐		Acqua Hotel	Views of Richardson Bay and Mount Tamalpais	Marin	User Rating: 4.6	9.2 mi	Check Rates

San Francisco Bay Area Lodging Highlights

San Francisco's hotels run the gamut from head-of-state luxury to modest motels. The City's premier hotels are almost all in the downtown area including Nob Hill, Union Square, and the Financial District/Embarcadero. Motels along the main corridors in and out of town, and pleasant bed-and-breakfasts in attractive, residential neighborhoods also exist. As a tourist destination with limited space, San Francisco hotels can charge premium prices, and do, especially downtown and on Nob Hill.

[Read More...](#)

Top Rated Places to Stay in San Francisco Bay Area

[Add to Trip](#)

Plan Your Trip

[Proceed to Trip Planner](#)

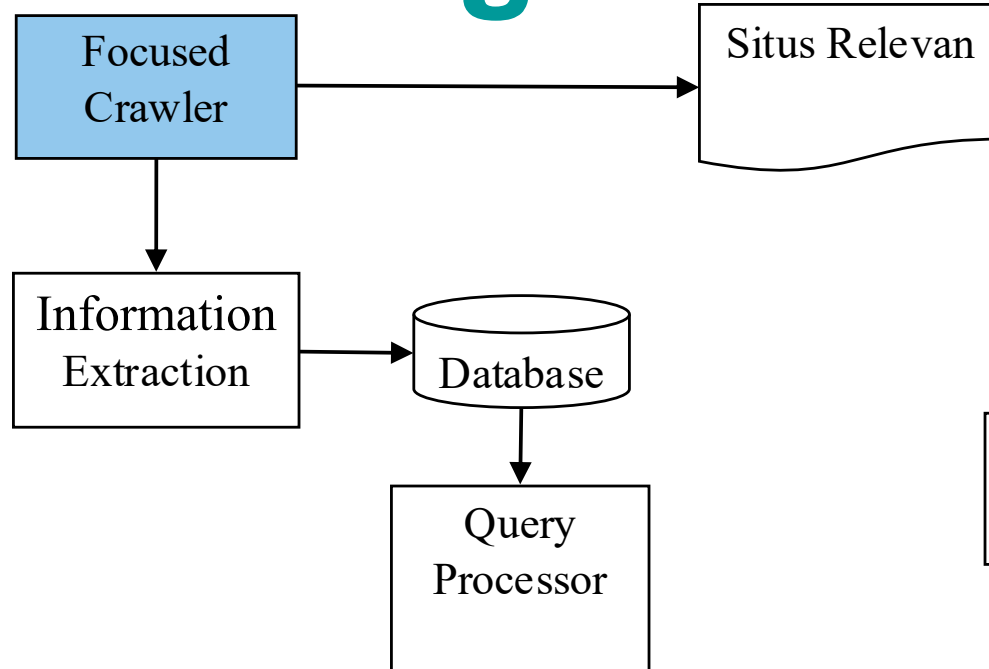
My Trip 2
[Create New](#) | [Change Trip](#)
May 22 - May 24
San Francisco Bay Area

- North Beach Restaurant
- 111 Minna Gallery
- 1801 Inn
- Mission Cliffs Rock Climbing Center
- Presidio Golf Course
- Masa's
- Hotel Rex

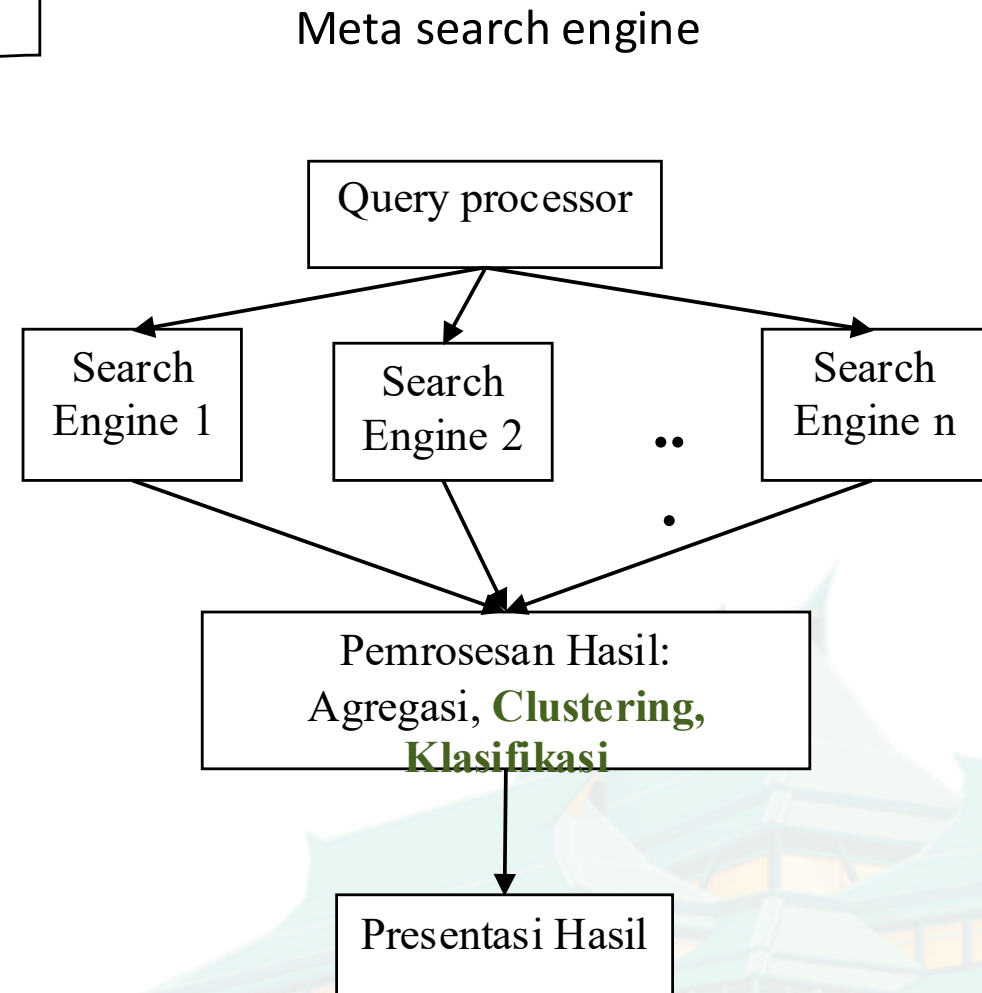
7 items in your trip



Search Engine: Architecture

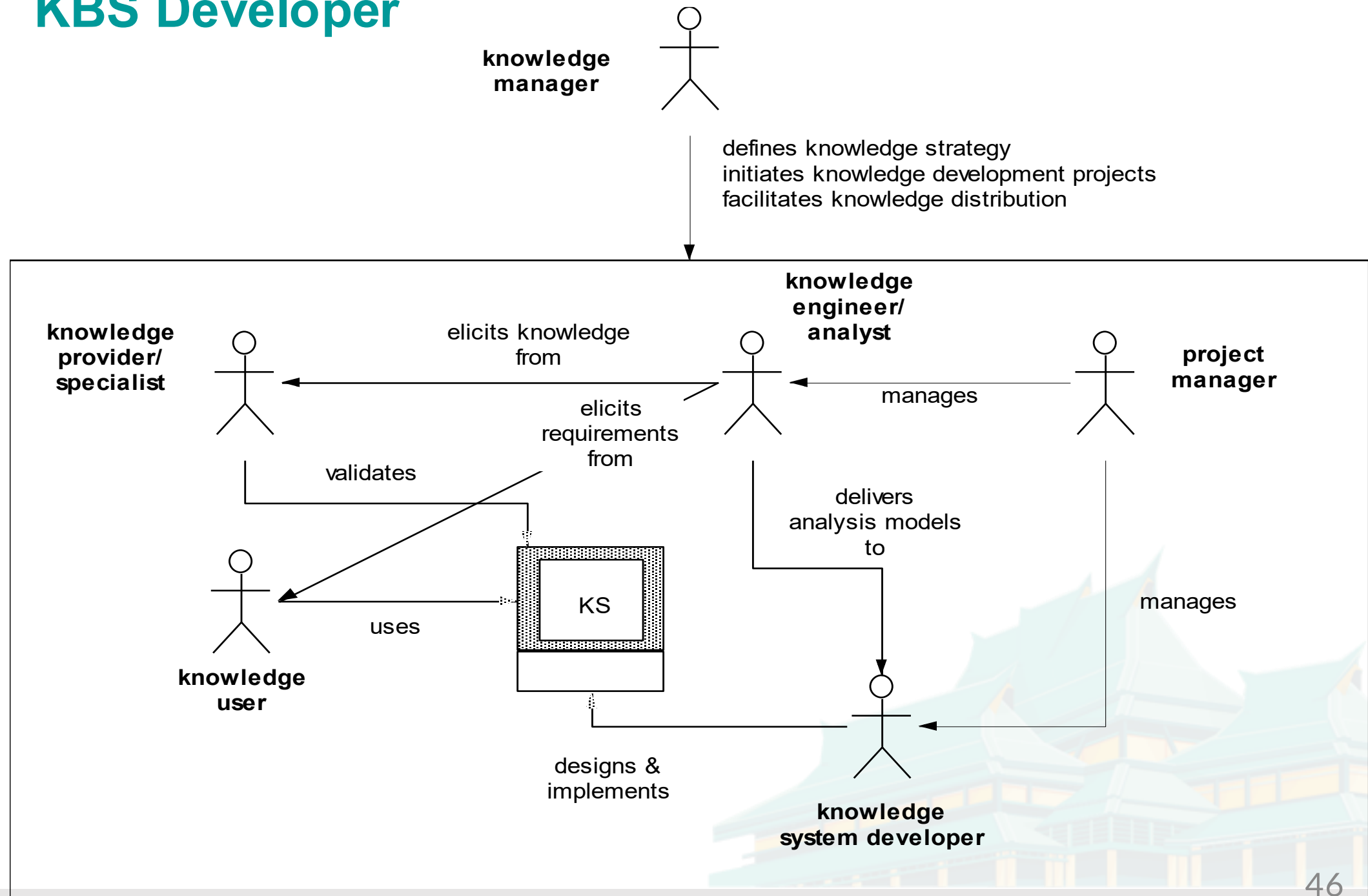


Vertical search engine





KBS Developer





Rekayasa Pengetahuan

- Akuisisi pengetahuan dalam suatu domain dari satu atau lebih sumber non-elektronik dan konversinya ke dalam suatu bentuk yang dapat digunakan oleh komputer untuk memecahkan persoalan yang umumnya hanya dapat dipecahkan oleh pakar domain tersebut.





Akuisisi Pengetahuan (KA)

- KA=knowledge *elicitation* + *representation*
- knowledge elicitation
 - Proses ekstraksi pengetahuan domain dan strategik dari pakar
 - Interview antara KE dan pakar
 - a cyclical process
- Knowledge representation
 - Proses merepresentasikan pengetahuan hasil ekstraksi ke suatu bentuk formal





Task dalam Knowledge Elicitation

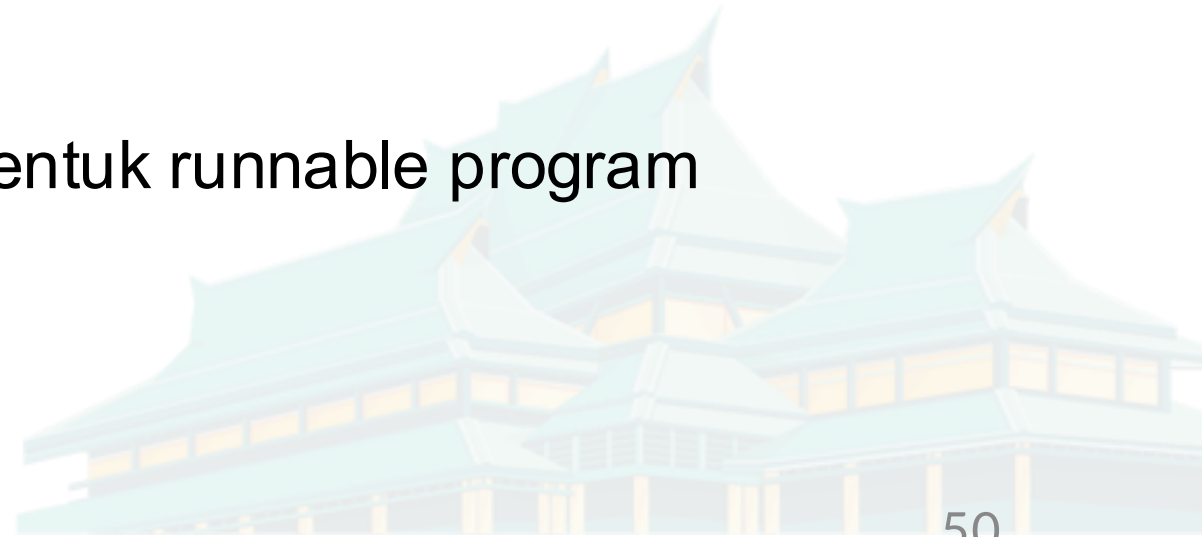
- Pada setiap iterasi:
 - **collect** knowledge (e.g. from expert)
 - determine **key concepts** in problem domain
 - establish **relationships** between various concepts in problem domain
 - decide **how knowledge is represented** in KBS
 - determine what knowledge needs to be collected in the next cycle





Tahapan Akuisisi Pengetahuan

- Identification
 - Identifikasi karakteristik masalah
- Conceptualization
 - Menemukan konsep2 untuk merepresentasikan pengetahuan
- Formalization
 - Design struktur untuk mengorganisasikan pengetahuan
- Implementation
 - Formulasi pengetahuan ke bentuk runnable program
- Testing
 - Validasi pengetahuan





Teknik Akuisisi Pengetahuan

- Manual:
 1. Interview
 2. Observasi
 3. Intuitive: tukar peran Knowledge Engineer dan pakar
- Otomatis:
 - Menggunakan tools untuk memfasilitasi akuisisi
 - Tools untuk pakar
 - Tools machine learning





Thank You

